CelebA		
Fraction Pruned	**Top 1**	**# Modal CIEs**
0	94.73	-
0.3	94.75	555
0.5	94.81	638
0.7	94.44	990
0.9	94.07	3229
0.95	93.39	5057
0.99	90.98	8754
Quantization	**Top 1**	**# Modal CIEs**
hybrid int8	94.65	404
fixed-point int8	94.65	414

Table 1: CelebA top-1 accuracy at all levels of pruning, averaged over runs. The task we consider for CelebA is a binary classification method. We consider exemplar level divergence and classify Compression Identified Exemplars as the examples where the modal label differs between a population of 30 compressed and non-compressed models. Note that the CelebA task is a binary classification task to predict whether the celebrity is blond or non-blond. Thus, there are only two classes. ***Note that the number of Taxicab CIEs are just the fraction of the threshold -ie if we threshold at 90% then the number of *CIE*s will be 10% of the dataset.

CelebA Top-1 Accuracy					
	Modal *CIE*s		**Taxicab *CIE*s**		
Fraction Pruned	**CIEs**	**All**	**90th**	**95th**	**99th**
30.0	49.82	94.75	63.58	58.49	55.35
50.0	50.55	94.81	63.06	58.88	54.44
70.0	52.61	94.44	64.08	61.36	55.29
90.0	50.41	94.07	62.35	56.60	50.10
95.0	45.57	93.39	60.53	51.99	43.43
99.0	39.84	90.98	49.93	39.75	29.21
Quantization					
hybrid int8	48.90	94.65	61.69	54.89	45.65
fixed-point int8	48.13	94.65	61.68	54.41	45.15

Table 2: A comparison of model performance on Compression Identified Exemplars (CIE) relative to performance on the test-set and a sample excluding *CIE*s (non-*CIE*s). Evaluation on *CIE* images alone yields substantially lower top-1 accuracy. Note that CelebA top-5 is not included as it is a binary classification problem.

et al., 2011; Jacob et al., 2018), with the first 100 training examples of each dataset as representative examples. Each of these quantization methods has open source code available. We use the MLIR implementation via TensorFlow Lite (Jacob et al., 2018; Lattner et al., 2020).

2.2 Results

Our baseline non-compressed model obtains 94.73% mean top-1 test-set accuracy (top-5 accuracy is not salient here as it is a binary classification task). Table 5 (top row) shows baseline error metrics across unitary and intersectional subgroups. There is a very narrow range of difference in overall test-set accuracy between this baseline and the different compression levels we consider. For example, after pruning 90% and 95% of network weights the top-1 test-set accuracy is 94.07% and 93.39% respectively. Table 2 provides details of performance at all compression levels for both pruning and quantization.

How does compression amplify existing model bias? We find that compression consistently amplifies the disparate treatment of underrepresented protected subgroups for all levels of compression that we consider. While aggregate performance metrics are only minimally affected by compression – albeit with FNR being amplified to a greater extent that FPR – we clearly see the newly introduced errors are unevenly distributed across sub-groups. For example,the

Model	Metric	Aggregate	Unitary				Intersectional			
			M	F	Y	O	MY	MO	FY	FO
Baseline (0% pruning)	Error	5.30%	2.37%	7.15%	5.17%	5.73%	2.28%	2.50%	5.17%	5.73%
	FPR	2.73%	0.93%	4.12%	2.59%	3.18%	0.81%	1.12%	2.59%	3.18%
	FNR	22.03%	62.65%	19.09%	21.35%	24.47%	60.45%	66.87%	21.35%	24.47%
Normalized Difference Between **1)** Compressed and **2)** Non-Compressed Baseline										
Compressed (95% pruning)	Error	24.63%	24.49%	24.67%	20.64%	35.84%	7.96%	49.12%	20.64%	35.84%
	FPR	12.72%	49.54%	6.32%	3.35%	36.02%	5.37%	101.88%	3.35%	36.02%
	FNR	34.22%	8.41%	40.30%	33.83%	35.39%	9.21%	6.98%	33.83%	35.39%

Table 3: Performance metrics disaggregated across `Male` (M), `not Male` (F), `Young` (Y), and `not Young` (O) subgroups. For all error rates reported, we average performance over 10 models. **Top Row**: Baseline error rates, **Bottom Row:** Relative change in error rate between baseline models and models pruned to 95% sparsity,

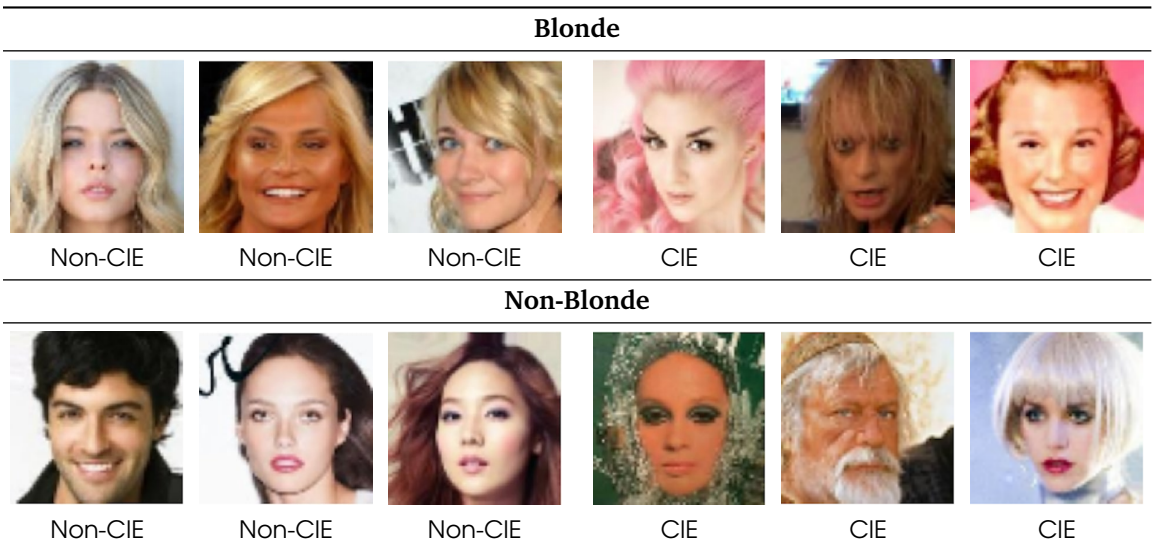

Figure 3: Compression Identified Exemplars (CIEs) are images where there is a high level of disagreement between the predictions of pruned and non-pruned models. Visualized are a sample of CelebA CIEs alongside a non-CIE image from the same class. Above each image pair is the true label. We train a ResNet-18 on CelebA to predict a binary task of whether the hair color is blond or non-blond.

middle row of Table 3 shows that at 95% pruning FPR for `Male` has a normalized increase of 49.54% relative to baseline. In contrast, there is far more minimal impact on `not Male` with a normalized relative increase of only 6.32%. This is less than the overall change in FPR (12.72%). We note that this appears closely tied to the overall representation in the dataset, with `Blond not Male` constituting 14% of the training set versus `Blond Male` with only 0.85%. Compression cannibalizes performance on low-frequency attributes in order to preserve overall performance. In Table 4 we show that higher levels of compression only further compound this disparate treatment.

3 Auditing Compressed Models in Limited Annotation Regimes

In the previous section, we established that compressed models amplify existing bias using traditional error metrics. However, the auditing process we used and conclusions we have drawn required the presence of labels for protected attributes. The availability of labels is often highly infeasible in real-world settings (Veale & Binns, 2017) because of the cost of data acquisition and privacy concerns associated with annotating protected attributes. In this section, we propose Compression Identified Exemplars (*CIE*s) as an auditing tool to surface a tractable subset of the data for further inspection or annotation by a domain expert. Identifying a small sample of examples that merit further human-in-the-loop annotation is often critical given the large scale size of modern datasets. *CIE*s are where the predictive behavior diverges between a population of independently trained compressed and non-compressed models.

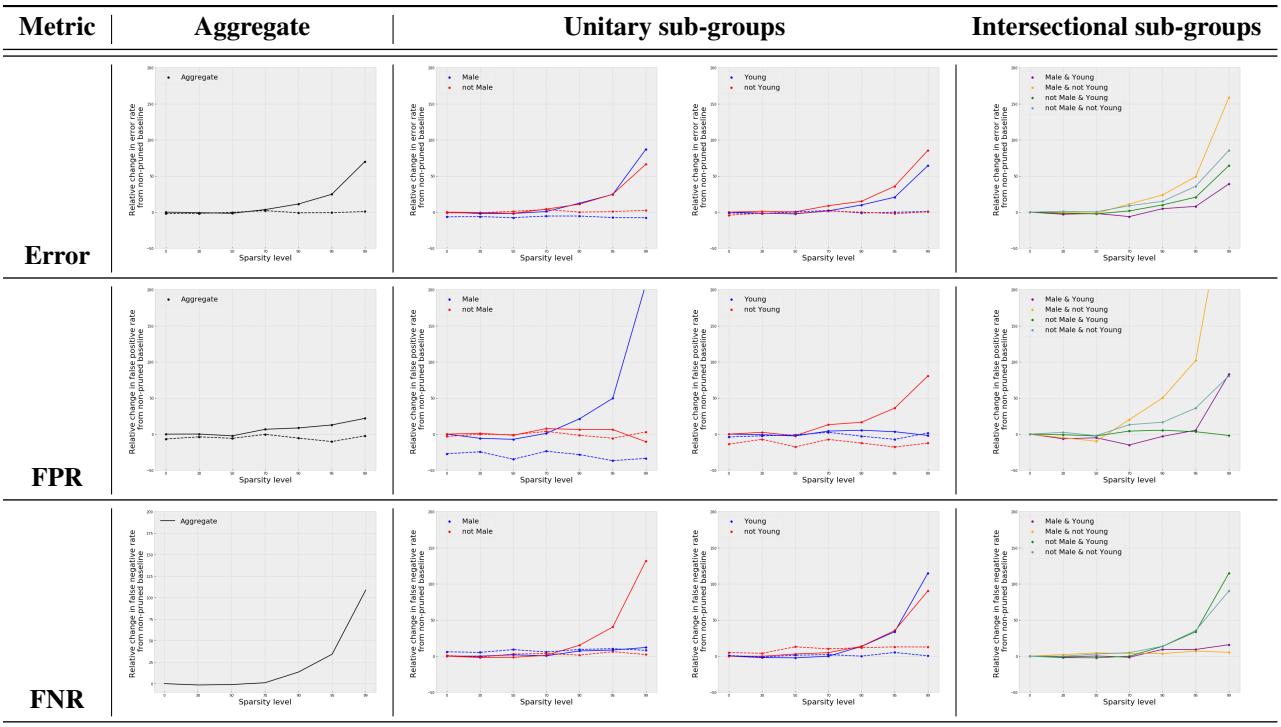

Table 4: For each unitary and intersectional sub-group, we plot the normalized difference of the compressed model, at each level of sparsity (x-axis), relative to the non-compressed model. Note that we threshold the y-axis limit at 100 for the purposes of standard comparison. **Top row:** Aggregate error, **Middle row:** False Positive Rate (FPR), **Bottom row:** False Negative Rate (FNR)

3.1 Divergence Measures

In additional to the measure of divergence proposed by proposed by Hooker et al. (2019b) which we term *Modal CIE*, we consider an additional measure of divergence *Taxicab CIE*. We briefly introduce both below. We provide a proof in the appendix of the equivalence of *CIE*-selection algorithms based on the Jaccard and Taxicab distances.

Modal *CIE* Hooker et al. (2019b) For set $Y^*_{x,t}$ we find the *modal label*, i.e. the class predicted most frequently by the t-compressed model population for exemplar x, which we denote $y^M_{x,t}$. Exemplar x is classified as a Modal CIE_t if and only if the modal label is different between the set of t-compressed models and the non-compressed models:

$$CIE_{x,t} = \begin{cases} 1 & \text{if } y^M_{x,0} \neq y^M_{x,t} \\ 0 & \text{otherwise} \end{cases}$$

Taxicab *CIE* We compute Taxicab distance as the absolute difference between the distribution of labels y^M_0 from the baseline models and the set y^M_t from the compressed models. Given an example x, define $B_x = \{b_{x,i}\}$ to be the distribution of labels from a set of baseline models where $b_{x,i}$ is the number of baseline models that label example x with class i. Similarly define $V_x = \{v_{x,i}\}$ to be the distribution of labels from a set of variant models where $v_{x,i}$ is the number of variant models that label example x with class i.

Let d_T be the Taxicab distance between two label distributions,

$$d_T(B_x, V_x) = \sum_i |b_{x,i} - v_{x,i}|.$$

Difference between measures proposed While *Modal CIE* identifies all examples with a changing median label as *CIE*, *Taxicab CIE* scores the entire dataset allowing for a ranking that can be thresholded by a domain user. Both methods of auditing require no labels for the underlying attributes. That said, note that this turns into a limitation in an overfit 0% training error regime as without any predictive difference it would not be possible to compute *CIE* using either measure in the training set.